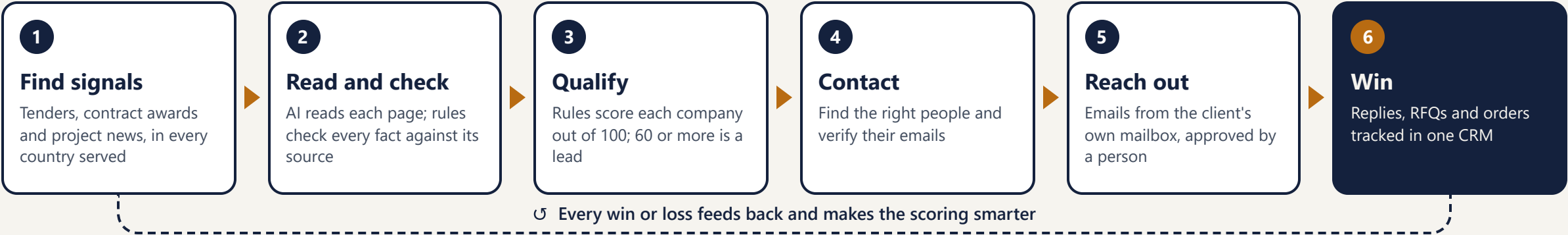


AI lead generation for pipeline and industrial projects

Assessment and build proposal · September 2026 · One-page summary; full detail and diagrams follow



Optional premium services can plug into steps 1, 3, 4 and 5 (project databases, trade records, contact data, sending tools) for even better leads.

2 weeks

Phase 0: fix security and measure accuracy

24 weeks

Full build of all steps, in six phases

\$130–365 / month

Running cost once live: Claude API usage (~\$100–300), database and server. AI coding tools for the build are separate.

85% → 90%

Leads correct, month 3 → month 12

Where the prototype stands

- Covers about a quarter of the target workflow
- Open database and no login: fixed first, in two weeks
- The good parts (interface, evidence rules) are kept

How we build it

- Free public sources in every target country, plus our own code
- Seven source criteria and a 100-point score decide who is a customer
- Claude AI reads; rules decide; every fact is linked to its source

What the client gets

- Qualified buyers with evidence, in any market
- Verified contacts and outreach from its own mailbox
- One CRM from first signal to order; it owns the data and code

Assessment and build proposal

Full detail · September 2026 · Diagrams appear on landscape pages next to the sections they belong to.

Summary

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Summary

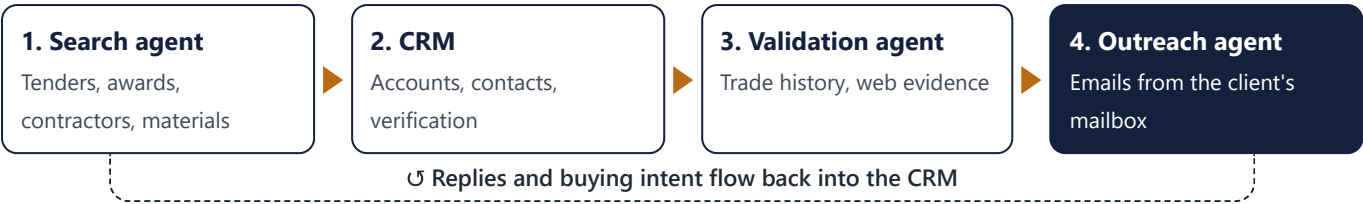
We recommend a 24-week build that turns the current prototype into an AI system that finds buyers for pipeline and industrial projects in every country the client serves, including the GCC, from public procurement signals, confirms their need and opens conversations from the client's own mailbox. It runs on free sources, open-source software we host ourselves, and tools the client already has. The only recurring cost is AI usage and hosting, about \$130–365 a month.

- **Where it stands:** the prototype is well organised and covers about a quarter of the target workflow. Search partly works, the CRM is mostly screens, buyer validation is a placeholder and outreach does not exist.
- **What must change first:** the database is open to anyone with the URL, there is no login, contact details can be attached to the wrong person, and the dashboard shows sample data. A two-week Phase 0 fixes all of this.
- **Free first:** search, page reading, contacts, background jobs and the CRM run on free sources, self-hosted open-source tools or the client's existing tools. Paid data is optional and is added only if a measured gap costs deals (section 5).
- **How we decide who is a customer:** seven criteria choose which websites to watch, and a 100-point score, calculated by rules, decides which companies qualify (section 6).
- **Rules decide, AI reads:** deterministic code collects, checks, matches, scores and enforces sending limits. Claude only reads text and writes, and code checks every AI answer (section 7).
- **A harness with premium plug-ins:** every step has a free in-house default and an optional premium provider behind the same interface, so the client can buy top-notch quality where it pays off, such as contact databases, Volza or project databases, without rebuilding anything (section 9).
- **More than contracts:** eleven buying signals, six buyer types and an eight-stage funnel from first signal to purchase order (section 8).
- **Benchmarks:** 85% or more of leads correct by month 3 and 90% by month 12, a reply rate at or above the 3.43% cold-email average, and a spam rate under Google's 0.3% limit (section 12).
- **Cost:** about \$130–365 a month to run, plus AI coding tools (Claude Code, Codex) at \$20–200 per developer per month during the build (section 15).

1. The target workflow

The client wants an AI sales team that finds industrial buyers before competitors do, proves they need the product, and starts the conversation from the client's own mailbox. There are four stages, and every stage writes back to one

CRM.



The loop back from stage 4 matters most: replies are the strongest intent signal the system will ever get.

Stage	What it must do	Output into the CRM
1. Search agent	Take a topic or project, find who won the tender, the EPC contractor and subcontractors, their key people, and the materials the project needs (pipes, steel, valves)	Project, owner, contractors, material packages, contacts, source evidence
2. CRM	Show everything found, and verify contact details	Verified contacts, stage, owner, activity history
3. Validation agent	Research each buyer's past purchases on the web and in Volza trade data; confirm they really need the suggested products	Need score with reasons: "imported 4,200 t of API 5L line pipe in the last 12 months"
4. Outreach agent	Send proposals from the client's mailbox, handle replies, test buying intent	Emails sent, replies, intent status updated automatically

2. Where the prototype stands today

The prototype covers roughly a quarter of the target workflow. Stage 1 works partly, stage 2 mostly as screens, stage 3 is a placeholder and stage 4 does not exist. We audited the full codebase (about 15,000 lines) on 24 Sep 2026: it type-checks cleanly, all 51 automated tests pass and the production build succeeds.

Stage	Built	What exists	What is missing
1. Search agent	~55%	Live search (Google via SerpAPI), page scraping (Firecrawl), AI extraction of company, project, contractor and material fields, source-quality filters, five search "plays"	It is a fixed pipeline of about 2,000 lines of keyword rules, not an agent. Each run lives inside one web request capped at 28 seconds, so only 6–12 pages are read. No scheduled watchlists, no tender-portal connectors and no local-language sources. Parts are hard-coded to a few sample queries and one region's buyer list, so it cannot be pointed at other countries without code changes.
2. CRM	~40%	Screens for candidates, leads, decision makers, lead lists, activities and CSV export; 13 database migrations	No login. Contact enrichment and email verification queues fill up but nothing processes them, so no email can ever reach "Verified". Data is split between the browser and the database. The dashboard shows hardcoded sample records, not real data.
3. Validation agent	~5%	A trade-data interface with a Volza placeholder that always returns "Pending"	Volza integration, purchase-history matching, need scoring, the agent itself
4. Outreach agent	0%	Nothing	Mailbox connection, sending-domain setup, sequences, AI drafting, reply handling, CRM auto-update, opt-out handling

What the FDE did well: clean module boundaries with swappable providers, AI output validated before use, an evidence-first rule (no invented emails, unknown stays unknown), transparent run logs showing why each source

was accepted or skipped, and honest client documentation of what is planned versus built. We keep all of this. What we replace is the engine underneath it and everything that is missing.

3. Issues that block real use

Three issues make the prototype unsafe to put in front of a sales team today: the database is open to anyone, contact data can be wrongly attached to people, and the dashboard is not real. All of them can be fixed in the first two weeks.

#	Issue	Evidence in the code	Impact	Severity
1	Anyone with the app URL can read, change or delete the whole CRM	<code>/api/discovery/crm</code> uses the database's master (service-role) key, which bypasses all access rules, and checks no login. <code>proxy.ts</code> enforces nothing and excludes <code>/api</code> .	Full leak of leads, contacts and notes; data can be wiped	Critical
2	No login	<code>/login</code> is a placeholder: "Auth wiring pending credentials"	No users, roles or audit trail	Critical
3	Master database key was shared in chat	Stated in the project's own setup notes	Must be rotated before any deployment	Critical
4	Contacts can get another person's LinkedIn profile or email	<code>nearestLinkedIn()</code> falls back to the first LinkedIn URL on the page; <code>nearestEmail()</code> gives a page's only email (often info@) to every person and labels it "source-backed"	Wrong person contacted; damages sender reputation and trust	High
5	Dashboard shows hardcoded sample data	<code>repository.ts</code> returns records from <code>verified-data.ts</code> and never reads the database	Management sees numbers that don't reflect real work	High
6	Two competing data models	Discovery writes flat <code>discovery_*</code> tables with text IDs; the structured company, signal and score tables are never used	No de-duplication; no project → owner → contractor → material graph	High
7	Buyer score is mostly fixed	Every company gets +5 "company fit" automatically; intent and relationship scores are always 0	Ranking is not trustworthy	Medium
8	Personal data collected without a compliance basis	Named people are pulled from Google results and LinkedIn snippets and stored with no consent, retention or opt-out handling	Exposure under data-protection laws in each market (for example UAE PDPL, Saudi PDPL, EU GDPR) and LinkedIn's terms	Medium
9	Unreproducible builds	Almost every dependency is pinned to <code>latest</code>	An upstream release can break the build without warning	Low

4. Positioning: procurement intent, not web-browsing intent

We recommend not cloning 6sense, Bombora or ZoomInfo. Their "intent" is built for software buyers researching online, and it barely sees an EPC contractor buying line pipe for a pipeline project. For industrial project buyers, buying intent is public and specific: a tender, a contract award, an import record, a reply to an email.

	6sense / Bombora / ZoomInfo	This platform
What "intent" means	An anonymous company reading more articles on a topic (Bombora's publisher network), website visits (6sense), plus a contact database (ZoomInfo)	A named project needs a named material, won by a named contractor, within a procurement window
Signal source	Mostly English-language business and tech websites	Tender portals, EPC award news, owner announcements, import and export records, email replies
Fit for industrial project buyers	Weak: procurement happens through portals and relationships, not web research	Strong: built around how industrial projects are procured, country by country

	6sense / Bombora / ZoomInfo	This platform
Evidence	A topic score with no source	Every claim links to a source page and the exact quote
Typical cost	Enterprise contracts, commonly tens of thousands of dollars a year per product	Mostly free public sources; we pay only for AI usage (see section 15)

The FDE's roadmap suggested adding all 12 layers of the 6sense stack, including technographics and website-visitor identification. For a steel and pipe supplier, those layers add cost without adding buyers. We keep two ideas from those products: an explainable account score, and a contact lookup that tries several sources in turn.

5. Free first: build in-house and use what the client already has

Almost every layer can run on free sources, open-source software we host ourselves, or tools the client already pays for. The only unavoidable usage cost is the AI model. Paid data stays optional and is added only if a measured gap is costing deals.

Layer	Free or in-house way	What the client may already have	Paid option, only if needed later
Finding news and projects	GDELT news API (free, no key), Google News and trade-press RSS feeds, owner newsrooms, Brave Search API within its \$5 monthly free credit (~1,000 searches)	None	Brave or Claude web search at \$5–10 per 1,000 searches
Reading web pages	Self-hosted Crawl4AI or Crawlee (both Apache 2.0) on our own server; Jina Reader free key	None	Firecrawl cloud
Tender portals	Our own reader for each portal	Supplier registrations	Tender booklets, bought only for priority tenders
Stock-exchange announcements	Our own reader for exchange announcement pages in each target market	None	None
Contacts	Company websites, press releases and annual reports; email format worked out from public addresses; self-hosted Reacher mail-server check; Hunter free plan (50 credits a month) for spot checks	None	Apollo from ~\$49 per user a month, if the fill rate is too low
Import history	UN Comtrade (free, country level) for market sizing	Volza account	Volza, if not already held
CRM	Zoho CRM Free (up to 3 users) or our own CRM screens	Existing CRM	None needed
Email sending	Gmail or Microsoft 365 API	Existing company email	Extra mailboxes on a sending domain
Background jobs	A Postgres job queue (Supabase Queues and <code>pg_cron</code>)	None	Inngest or Trigger.dev
Sanctions screening	Free OFAC , UN and UAE lists	None	None
AI	Claude API, pay per use; the Batch API halves the price of scheduled runs	None	None

Two free options we deliberately avoid. Google's Custom Search API is [closed to new customers](#) and shuts down on 1 Jan 2027. The free Gemini API tier allows Google to [use prompts to improve its products](#) for UAE users, so we don't send client data there.

Can we get intent data from CDN or internet providers?

No. CDNs only process traffic for their own customers and state that they do not use or sell end-user data for marketing ([AWS](#), [Cloudflare](#), [Fastly](#)). Buying it would also breach data-protection law: the [UAE PDPL](#) treats online

identifiers such as IP addresses as personal data, with consent as the default basis, and the [Saudi PDPL](#) requires consent or prior interaction for direct marketing. The EU GDPR also treats IP addresses as personal data.

The established vendors get intent data like this:

- **Bombora:** a [co-op](#) of thousands of B2B websites that run Bombora's tag and share consented reading data.
- **6sense:** its own network of B2B web activity, keyword tracking in more than 40 languages, and partner feeds including Bombora, G2 and TrustRadius ([6sense](#)).
- **ZoomInfo:** a contact database built over many years from licensed, contributed and researched data.

For this client, the detail comes from better sources:

- **Tender documents.** Many government portals share or sell full tender documents containing the bills of quantities, with pipe sizes, grades and quantities. Saudi Etimad's rules are one example ([executive regulations](#), articles 33 and 69).
- **Import records.** Volza shows each shipment's product code, quantity, supplier and date.
- **Award chains.** An EPC contract award is usually followed, within weeks to a few months, by bulk orders for line pipe, plates and valves. The platform tracks that chain from award to order.
- **The client's own signals.** Website visits, email opens, clicks and replies, RFQ forms, and meetings at industry trade shows in each market.

6. How we decide who is a customer

A company becomes a lead only when a written checklist says so. First we choose which websites are worth watching, using seven criteria. Then every company found there is scored against the client's customer rules. The rules are plain, visible and editable by the sales team.

Which websites we watch

Criterion	Rule
Relevant	Publishes projects, tenders, contract awards or procurement news in the client's sectors and target countries
First-hand	Official or primary sources (government portals, project owners, listed companies) rank above republished news
Fresh	Updates at least weekly
Open	Public page, RSS feed or API; respects robots.txt and the site's terms; never bypasses a login
Readable	Stable HTML, RSS or PDF that we can parse reliably
Language	English and the local languages we support
Proven	Keeps its place only if its items turn into qualified leads, measured monthly

Source category	Examples	How we read it	Tier
Government tender portals	The national e-procurement portal of each target country, e.g. Saudi Etimad, EU TED, US SAM.gov, UK Find a Tender; several offer free APIs	A reader built for each portal, daily	A
Project owners' newsrooms	Oil, gas, water and power companies, pipeline operators and utilities in each market	RSS or page crawl, daily	A
Stock-exchange announcements	Contract awards announced by listed owners and contractors, e.g. Saudi Tadawul, UK regulatory news, US SEC filings	Announcement feeds, daily	A
Trade press	Pipeline, oil and gas, water and construction trade publications	RSS headlines, then the article	B
Global news index	GDELT	Free API, several times a day	B

Source category	Examples	How we read it	Tier
Contractor websites and careers pages	EPCs and fabricators already in the buyer graph	Crawl weekly	B
Trade records	Volza (client's account), UN Comtrade (country level)	Export or API	A

The source list is set up per country. Adding a new market means adding its portals, exchanges and news sources to the list, not changing code.

Which companies count as customers (score out of 100)

Rule	What it checks	Points
Product fit	The project or company needs a product family the client sells, matched by keywords and customs (HS) codes	0–25
Timing	The signal is inside the buying window: open tender, award in the last 9 months, RFQ	0–20
Buying role	Owner, EPC contractor, subcontractor, fabricator or stockist, weighted by how often each buys directly	0–15
Market	Located in, or delivering to, the countries the client serves	0–10
Size	Project value or yearly import volume above a threshold the client sets	0–10
Evidence	Official source, exact quote stored, confirmed by a second source	0–10
Reachable	At least one verified decision maker	0–10

Hard stops (never scored): existing customers (sent to their account owner), competitors, sanctioned companies (checked against the free UN and US OFAC lists), completed or cancelled projects, and markets the client doesn't serve.

A company scoring 60 or more becomes a qualified lead. The threshold and weights start as agreed defaults and are tuned from the client's won and lost deals. AI reads the pages and fills in the facts, but the score is always calculated by these rules, so the same facts always give the same score.

7. How we build it in-house: rules decide, AI reads

Every step runs deterministic code first and uses AI only where language has to be understood. Code is free to run, repeatable and auditable. AI is paid per use, so it is reserved for reading unstructured text and writing, and code checks every AI answer before it is saved.

Step	Deterministic code and rules	AI (Claude)	Free tools
1. Collect	Source list, schedules, RSS and API readers, portal readers, de-duplication by URL and content fingerprint	None	Crawl4AI or Playwright, feedparser
2. Filter	Keyword, region, date and source-tier rules drop most pages	A low-cost model (Claude Haiku) only for pages the rules can't decide	None
3. Extract	Parse dates, values, reference numbers, tables and PDF text	Reads articles and tender text to pull out companies, roles, scope and materials, each with its quote	PDF text extractors
4. Check	The quote must appear word-for-word on the page; values must be from allowed lists; dates and numbers must parse	None	Schema validation
5. Match companies	Name normalisation, alias table, fuzzy matching in Postgres, website-domain match	Tie-breaker for unclear names, e.g. the same company written in two languages	Postgres trigram search

Step	Deterministic code and rules	AI (Claude)	Free tools
6. Score	The customer rules in section 6	Writes the plain-language reason	None
7. Find contacts	Crawl leadership pages, team pages and press releases; work out the email format from public addresses; check the mail server	Maps job titles to buying roles	Reacher (self-hosted email checker)
8. Validate need	Match customs codes in import records; count past awards	Summarises the evidence for the salesperson	Volza (client's account), UN Comtrade
9. Outreach	Templates, daily sending limits, opt-out and suppression list, send windows	Drafts the personal opening lines; sorts replies	Gmail or Microsoft 365 API
10. Learn	Conversion rate per source and per signal; rule weights adjusted	Suggests new rules from patterns; a person approves them	None

Over time the split moves further towards code. When the AI keeps making the same judgement, we turn it into a rule. That lowers the cost per lead and makes results more consistent.

8. Beyond contract awards: the full buyer funnel

Contract awards are one buying signal out of about ten. The platform follows a project from its first budget line to the material order, finds buyers who never appear in a tender, and moves every account through an eight-stage funnel.

Buying signals, earliest first

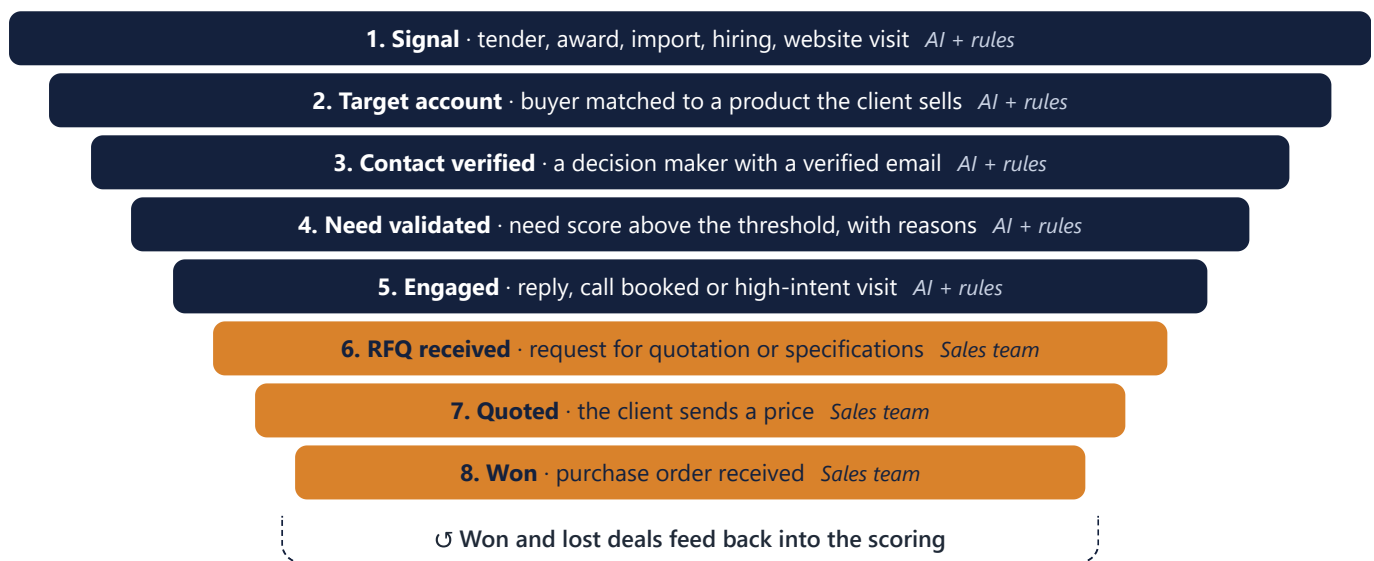
Signal	Example	Source	Approximate time before material purchase
Capex plan or budget	An owner adds a pipeline expansion to its five-year plan	Annual reports, government budgets, news	18–36 months
Project announced or design (FEED) contract awarded	Front-end engineering design contract signed	News, project databases	12–24 months
Permits and approvals	Environmental approval or land allocation granted	Regulator notices	12–18 months
Tender or prequalification	EPC tender issued; bidders prequalified	Tender portals	6–12 months
EPC contract award	Owner names the winning EPC contractor	Press releases, portals	1–9 months (peak window)
Subcontract or package award	Piping or structural package let to a subcontractor	News, contractor announcements	1–6 months
Hiring	Contractor hires piping or procurement staff for the project	Job boards	3–9 months
Material RFQ or vendor registration	EPC opens vendor registration or sends RFQs	Contractor vendor portals, email	0–3 months (hot)
Import history	A company imported line pipe every quarter last year	Customs records (Volza)	Recurring
Maintenance shutdown	A refinery schedules a turnaround	Operator notices, news	3–6 months
Engagement	Visits the client's website, replies to an email, meets at a trade show	Client's own website, mailbox and CRM	Now

Timings are indicative and vary with project size. The earlier the signal, the more time to get onto the approved vendor list before the order is placed.

Buyers who are not contract winners

Buyer type	What they buy	How the platform finds them
Project owners (national oil companies, utilities)	Materials they procure directly and supply to contractors	Capex plans, tenders
EPC contractors	Bulk materials for awarded projects	Contract awards
Subcontractors and fabricators	Piping spools, structural steel, packages	Package awards, hiring
Stockists, distributors, traders	Regular restocking for resale	Import records
Operating plants (maintenance)	Spares for shutdowns and repairs	Shutdown schedules, annual rate contracts
Equipment makers	Plate and pipe for tanks, skids, pressure vessels	Import records, product catalogues

The funnel



The agents run stages 1–5; the sales team owns 6–8, and every won or lost deal feeds back into scoring.

Stage	Moves forward when	Done by
1. Signal	A buying signal from the table above is captured with its source	Discovery agent
2. Target account	The buyer company is identified and matched to a product the client sells	Discovery agent
3. Contact verified	At least one decision maker has a verified email	In-house contact finder
4. Need validated	Need score passes the threshold, with written reasons	Validation agent
5. Engaged	The buyer replies, books a call or visits a high-intent page	Outreach agent
6. RFQ received	The buyer sends a request for quotation or specifications	Sales team, logged automatically from email
7. Quoted	The client sends a price	Sales team
8. Won	Purchase order received	Sales team

The dashboard shows conversion between every stage and which signal types produce orders, so budget moves to the signals that work.

9. Target architecture: an in-house harness with premium plug-ins

The platform is a harness. Our own code runs every step, each step has a free in-house default, and each step can take an optional premium provider through the same interface. The in-house version is good on its own; premium plug-ins are how the client buys top-notch lead quality in specific places, only when a measured gap justifies it.

The three diagrams at the end of this section show how the platform works, the system architecture, and the free version against real premium services.

Layer	Choice	Why
App and UI	Keep Next.js, React, TypeScript, Tailwind	The prototype's UI is solid; rebuilding it wastes time
Database	Keep Supabase Postgres; one schema; row-level security; trigram search for matching company names	Already in place; the unused structured tables from migration 0001 become the base of the graph
Background jobs	A Postgres job queue (Supabase Queues and pg_cron)	Durable multi-step runs, retries and schedules with no extra paid service
AI models	Claude Haiku 4.5 for filtering, Claude Sonnet 5 for extraction, Claude Opus 5 for qualification	Strong structured output; used only where rules can't do the job; the provider interface stays swappable
Search and page reading	GDELT, RSS feeds and self-hosted Crawl4AI behind interfaces; our own readers for tender portals and exchange announcements in each target country	Free, and the portals are higher signal than general web search
Contacts	In-house: company sites and press releases, email format from public addresses, self-hosted mail-server check	Free and owned; a paid source is added only if a measured gap costs deals
Trade data	The client's Volza account, if they have one; UN Comtrade for country totals	Past imports by customs code prove a buyer uses the product
Email	Microsoft 365 or Google Workspace API with OAuth, on a separate sending domain	Emails come from real mailboxes with proper authentication
Observability	Sentry free tier for errors, a job dashboard, cost logged per run	Know what failed and what each lead cost

In-house versus premium plug-ins, layer by layer

Layer	In-house default	Premium plug-in	Quality uplift	Cost band	Switch on when
Signal discovery	GDELT, RSS, newsroom, portal and exchange readers	Paid project databases covering the client's markets	Earlier stages (concept, design), subcontract packages, project contacts	Medium	Too few early-stage projects in the pipeline
Page reading	Crawl4AI on our own server	Firecrawl cloud	JS-heavy and bot-protected sites; less maintenance	Low	More than 10% of priority sources fail to read
Material detail	Public tender notices	Paid tender booklets	Exact sizes, grades and tonnage	Per tender	A priority tender in the client's product range
Company data	Own buyer graph and sanctions lists	Apollo or ZoomInfo company data	Size, parent company, web domains, industry codes	Low to high	Missing domains or bad matches block contact search
Contacts	Websites, press releases, email format and Reacher check	Apollo, then Lusha, Cognism or ZoomInfo; ZeroBounce	More mid-level buyers, direct dials, verified emails	Low to high	Verified contact on under 50% of priority accounts

Layer	In-house default	Premium plug-in	Quality uplift	Cost band	Switch on when
Need validation	Client's Volza, UN Comtrade, past awards	Volza full plan or API	Company-level import history per customs code	Medium	Qualified leads are not turning into RFQs
Intent timing	Own procurement signals and engagement	Bombora or 6sense	Research-surge timing (thin coverage for industrial buyers in many countries)	High	Only after a coverage test on the top 200 accounts
Outreach	Gmail or Microsoft 365 API, limits, approval	Instantly or Smartlead, extra sending domains	Warm-up, inbox rotation, deliverability at scale	Low	Sending more than about 50 emails per mailbox per day

Cost bands: low is under 100 US dollars a month, medium is 100–1,000, high is over 1,000 (checked September 2026). Each plug-in is trialled for 4–6 weeks on the same accounts and kept only if qualified leads, replies or RFQs rise. Suggested order: email verification, contact database, Volza, project database, then intent data.

How the agents avoid making things up:

- **Quote-backed facts.** Every extracted field stores the exact source sentence. A field with no quote is not saved.
- **A test set of real awards.** About 150 known contract awards and tenders from the client's markets with verified answers. Every change to prompts or code is scored against it before release.
- **Merging duplicates.** "Acme Pipelines Ltd", "Acme Pipeline Co." and the same name in a local script resolve to one company record.
- **People approve before sending.** In the first release the outreach agent drafts and a person approves; auto-send is unlocked per campaign once quality is proven.
- **Spending limits.** Each run has a cap on searches, pages read and AI spend.

How the platform works

One platform, five steps. It works on free sources, and premium data can be plugged in for even better leads.



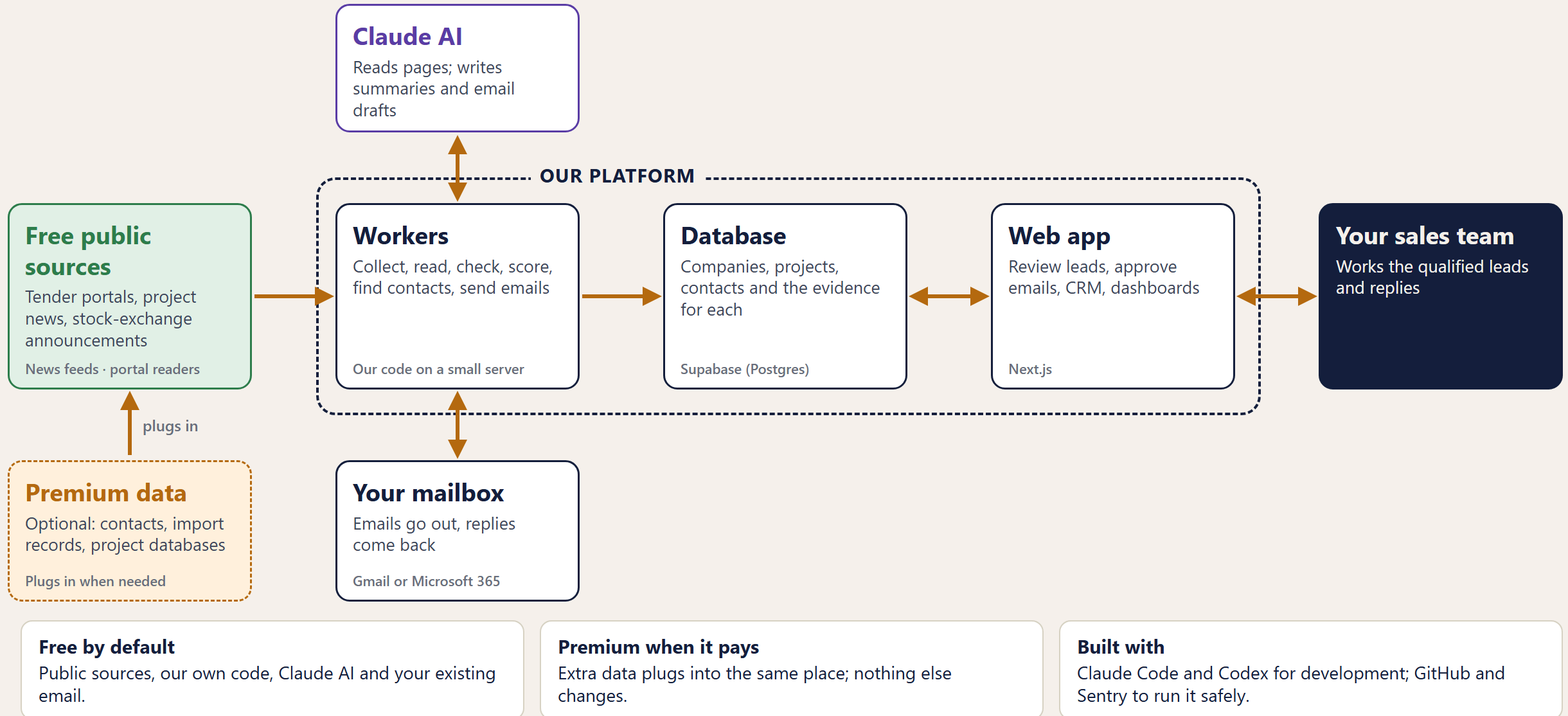
Free version
Good leads from public sources. Runs for about \$130–365 a month.

Premium add-ons
Better leads where public data is thin. Switch on one at a time.

Same platform either way
Adding premium data is a setting, not a rebuild.

System architecture

What we build and run (inside the dashed line), and the outside services it connects to.



Free version versus premium services

Every step works for free. These real services can be added, one at a time, to raise lead quality where it pays off.

Step	Free (built in)	Premium services you can add	What you gain	Cost
Find projects and tenders	News feeds, government tender portals, company and stock-exchange announcements	GlobalData , IJGlobal , Rystad Energy (project databases) · Tendersinfo , GlobalTenders (tender aggregators)	Projects found months earlier, at planning and design stage; named contractors and subcontractors; project values; one feed for many countries	\$\$-\$\$\$
Read hard websites	Our own page reader (Crawl4AI) + Claude AI	Firecrawl , Zyte , Bright Data , Apify	Pages behind JavaScript or bot protection read reliably; fewer failed sources; less upkeep	\$-\$
Prove they buy	UN Comtrade country totals, past contract awards	Volza , ImportGenius , Panjiva (S&P Global) trade records	Each company's actual shipments by product code: volumes, prices and current suppliers. Real buyers, their size, and who you would replace	\$-\$\$\$
Know the company	Our own records, public filings, free sanctions lists	Dun & Bradstreet , Orbis (Moody's)	Size, revenue, parent and subsidiaries, credit risk: target the entity that decides, avoid slow payers	\$\$\$
Spot the timing	Tender and award dates; visits and replies from the client's own website and emails	TheirStack , Coresignal (hiring data) · Bombora , 6sense (intent data)	Hiring for procurement or project roles shows a project starting; intent shows research surges (thin for industrial buyers, so test first)	\$-\$\$\$
Find and verify contacts	Company websites, press releases, email-format check (Reacher)	Apollo , Lusha , Cognism , ZoomInfo · Clay or FullEnrich (search many at once) · ZeroBounce , NeverBounce	More decision makers, including mid-level buyers; direct phone numbers; every email verified, so bounces stay under 2%	\$-\$\$\$
Reach out at scale	The client's own mailbox (Gmail or Microsoft 365), Google Postmaster Tools	Instantly , Smartlead , lemlist · GlockApps (inbox tests)	Warm-up, rotating inboxes, A/B tests, LinkedIn steps: more emails reach the inbox and get replies	\$

Start free, add one at a time

Add a service only where a measured gap costs deals. Trial it for 4–6 weeks; keep it only if leads, replies or RFQs rise.

Suggested order

Email verification, then contact data, then trade data, then a project database. Intent data last, after a coverage test.

Cost bands (indicative)

\$ under \$100 a month · \$\$ \$100–1,000 · \$\$\$ over \$1,000. Many are annual contracts; get quotes.

10. CRM: build our own or connect to an existing one

We recommend a hybrid: keep a focused intelligence workspace in this app (research, evidence, approvals), and sync qualified accounts and contacts to the client's existing CRM if they have one. That avoids spending budget rebuilding pipeline, reporting and mobile features that HubSpot or Zoho already do well.

Option	Best when	Pros	Cons	Effect on build
A. Full CRM in this app	The client has no CRM, or a small team (under ~10 sales users)	One system; full control of the agent workflow	We must build and maintain pipeline, tasks, reporting, permissions, mobile	About +6 weeks
B. Intelligence app + sync to HubSpot, Zoho or Salesforce (recommended)	The client already uses a CRM, or plans to	Salespeople stay in a familiar tool; mature reporting; less to build	Two-way sync work; per-seat CRM licence cost	About +3 weeks for one CRM connector
C. Agents only, no app of our own	A large team with a well-run CRM	Cheapest	Evidence and approval screens are squeezed into CRM fields; weak review experience	About -4 weeks

The roadmap assumes option B. A small team can use Zoho CRM's free edition (up to 3 users) or HubSpot's free CRM (2 users, 1,000 contacts) at no licence cost. If the client has no CRM, option A adds about 6 weeks.

11. Roadmap and timeline

The standard plan delivers the full four-stage workflow in 24 weeks, in six phases. Each phase ends with something the sales team can use and a measurable acceptance test. Phases overlap by one to two weeks.

Phase	Weeks	Delivers	Accepted when
0. Secure and stabilise	1–2	Login with Microsoft or Google accounts, roles, locked-down APIs, rotated keys, one database schema, contact bugs fixed, real dashboard, test set of ~150 known contract awards	No API responds without login; the dashboard matches the database; baseline accuracy measured
1. Discovery agent	3–8	Background agent runs, scheduled watchlists ("line pipe, UAE, weekly"), tender-portal and project-news connectors, quote-backed extraction, duplicate merging, new-lead alerts	≥85% of saved leads correct on the test set; ≥40 new qualified project/contractor accounts per week on the client's chosen watchlists
2. Contacts and CRM	7–12	In-house contact finder, email verification, CRM sync (or full CRM, option A), lead lists, export	≥70% of priority accounts have at least one verified decision maker; estimated bounce rate under 3%
3. Validation agent	11–15	Volza integration, product-to-HS-code mapping (e.g. 7304/7305/7306 for steel pipe), purchase-history matching, need score with written reasons	Every priority account shows a need score with sources; the client's sales lead agrees with ≥80% of the top-50 rankings
4. Outreach agent	14–20	Mailbox connection, sending domains with SPF, DKIM and DMARC, warm-up, sequences, AI-drafted emails with human approval, reply sorting (interested, not now, wrong person, out of office, unsubscribe), CRM auto-update, opt-out register	Bounce under 2%, spam complaints under 0.3%, 100% of replies sorted and logged within 5 minutes
5. Hardening and handover	19–24	Monitoring, cost dashboard, audit log, backups, PDPL documentation, load testing, training, runbooks	Two weeks of unattended operation with no critical incident

Faster and larger variants: a Lean MVP (phases 0–2, with a simpler validation step and outreach drafted by AI but sent by people) takes about 12 weeks. An Enterprise build (adds more countries, local-language sources and tender portals, analytics warehouse, win/loss learning for scoring, mobile) takes about 34 weeks.

12. How it changes over time, and benchmarks

The system starts rule-heavy, with people reviewing every lead. It gets better every month as the test set grows and won and lost deals retune the rules. We measure it against published benchmarks and against what the existing tools deliver.

Period	What the system does	People's role	Targets we measure against
Months 1–3	Sources and rules live; AI reads pages; buyer graph fills up	Review every lead	85% or more of saved leads correct on the test set
Months 3–6	Contact finder and outreach with approval; Volza checks if the client has an account	Approve every email	Verified contact on 50% or more of priority accounts; bounce rate under 2%
Months 6–12	Scores tuned on outcomes; weak sources dropped; repeated AI judgements become rules	Approve new rules; spot-check leads	90% or more correct; reply rate at or above the 3.43% cold-email average
Month 12 onwards	More countries, local-language sources and portals; auto-send on proven campaigns	Handle RFQs and quotes	Verified contact on 70% or more of priority accounts; AI cost per lead falling quarter on quarter

Targets are what we measure against, not guarantees. Phase 0 sets the real starting baseline.

Compared with the existing tools

Measure	6sense, ZoomInfo, Bombora	Our platform
Tenders, awards, bills of quantities, imports	Not covered	Core signals
Evidence behind a lead	A topic score or a database record	Source link and the exact quote
Contact coverage in the client's markets	Strongest for large enterprises	Senior contacts from public sources at first; grows with every reply
Outreach	Separate tools	Built in, from the client's mailbox
Ownership	Licensed; access ends with the subscription	The client owns the data, rules and code
Yearly subscription	About \$113k–126k for all three (Vendr medians and entry prices)	None; only tools and AI usage (section 15)

External benchmarks we hold ourselves to

- **Reply rate:** the average cold-email reply rate in 2025 was 3.43%, and the top quarter of senders got 5.5% or more ([Instantly 2026 benchmark report](#)).
- **Deliverability:** Google's sender rules require SPF, DKIM and DMARC, one-click unsubscribe, and a spam rate below 0.30%, ideally below 0.10% ([Google](#)). Yahoo also requires unsubscribes to be honoured within 2 days ([Yahoo](#)).

13. Is it practical?

Yes for finding projects, buyers and senior contacts from free sources. Only partly for exact material quantities and verified emails, and not at all for company-level import data without the client's Volza account. The table is the honest picture.

Capability	Possible with free sources and in-house code?	Notes
Find tenders, projects and contract awards in the client's markets	Yes	Public portals, owner newsrooms, exchange announcements, trade press
Name the EPC contractors	Yes	Awards are usually announced
Name subcontractors	Partly	Published less often; hiring and vendor-registration signals fill gaps
Exact material quantities (bills of quantities)	Partly	Inside tender booklets; needs the client's supplier registration, and some booklets carry a fee
Senior decision makers (directors, heads of procurement)	Yes	Company sites, press releases, annual reports
Mid-level buyers and package engineers	Partly	Less public; found through tender contacts and replies over time
Verified work emails	Partly	Email format plus a mail-server check works for many companies; domains that accept every address can't be verified
Company-level import history	Only with Volza	No free company-level source in most countries; UN Comtrade gives country totals only
Web-research intent (Bombora-style)	No	Not needed at the start; procurement signals are stronger here
Identify companies visiting the client's website	Partly	Works for office networks; most mobile traffic can't be identified
Automated outreach from the client's mailbox	Yes	Within daily sending limits, opt-out on every email and data-protection rules

What we need from the client

- A product list with customs (HS) codes, target countries and excluded accounts (current customers, competitors)
- Supplier registrations on the key tender portals, or permission to register on their behalf
- Login access to the tools they already pay for, such as Volza and their CRM
- Access to a mailbox and a separate sending domain for outreach, plus a named approver for emails
- A sales lead for about two hours a week to review leads and mark wins and losses

14. Team

Delivery is led by a senior AI engineer who owns agent quality, supported by a small cross-functional team.

Role	Owns
Tech lead / senior AI engineer	Agent design, prompts, the test set, data graph, architecture decisions
Senior full-stack engineer	CRM screens, login and roles, CRM sync, approvals and review UI
Data and integrations engineer	Tender portals, Volza, contact and verification APIs, email infrastructure
QA and data reviewer	Test-set labelling, weekly accuracy reviews, regression tests
Product / delivery manager	Weekly progress reviews with the client, priorities, acceptance sign-off
Designer	Review, approval and dashboard screens

The client provides one sales lead for about two hours a week, to label leads and approve outreach tone. Their feedback is what tunes the need score.

15. Development and running cost: tools and APIs only

Once live, the system costs about \$130–365 a month to run, mostly AI usage and hosting. During the build, each developer needs an AI coding subscription of \$20–200 a month. Everything else uses free tiers, open-source software, or tools the client already has.

Tools needed during the build (per developer, per month)

Tool	What it is for	Plan	USD/month
Claude Code	AI coding agent for building, testing and refactoring	Pro \$20; Max \$100 or \$200 for heavy daily use	20–200
OpenAI Codex	Second AI coding agent, e.g. for code review (optional)	Included in ChatGPT Plus \$20 or Pro \$100–200	0–200
GitHub	Code hosting, reviews, automated checks	Free plan	0
Supabase (development)	Test database	Free plan, 2 projects	0
Claude API (testing)	Running the test set and trial runs	Pay per use	30–100 per team

Example: three developers on Claude Max at \$100, plus one ChatGPT Plus seat for Codex reviews, cost about \$320 a month during the build, plus test API usage.

Running cost once live (per month)

Item	Choice	USD/month
AI models	Claude Haiku 4.5 (\$1/\$5 per million input/output tokens) for filtering, Sonnet 5 (\$2/\$10) for extraction, Opus 5 (\$5/\$25) for qualification; Batch API halves scheduled runs	100–300
Database	Supabase Pro (the free plan pauses after 7 days idle, so it is not for production)	25
Server for crawlers and workers	Hetzner CX23 (€5.49) or a slightly larger machine	7–20
App hosting	The same server, or Vercel Pro (the free Hobby plan forbids commercial use)	0–20
Web search and page reading	GDELT, RSS feeds, self-hosted Crawl4AI, Brave within its free credit	0
Contacts and email checks	In-house finder, self-hosted Reacher, Hunter free plan	0
Email sending, trade data, CRM	The client's existing email, Volza and CRM, or Zoho CRM Free	0 new
Monitoring	Sentry free plan	0
Total		~130–365

Assumes about 2,000 accounts researched and 3,000 emails a month; AI cost grows with volume. Occasional one-off costs: tender booklets for priority tenders, and a sending domain (about \$10–15 a year). Prices checked in September 2026.

What AI costs, and how it is billed

Two different kinds of AI cost. Neither includes premium data services or developer salaries.

DURING THE BUILD

AI for development

Fixed monthly subscription per developer. Stops or drops after the build.

Tool	Plan	USD / month
Claude Code	Claude Pro, or Max for heavy daily use	20 · 100 · 200
OpenAI Codex (optional)	Included in ChatGPT Plus or Pro	20 · 100–200
Claude API for testing	Pay per use, whole team	30–100

Example: 3 developers on Claude Max at \$100, plus one ChatGPT Plus seat for reviews, is about **\$320 a month**, plus test usage.

ONCE LIVE

AI for analysis inside the product

Claude API, pay per use: billed per million tokens read (input) and written (output).

Model	Used for	Input / output, per million tokens
Claude Haiku 4.5	Filtering pages	\$1 / \$5
Claude Sonnet 5	Extracting companies, drafting emails	\$2 / \$10
Claude Opus 5	Qualifying buyers	\$5 / \$25

Scheduled runs through the Batch API cost half. At about 2,000 companies researched and 3,000 emails a month: **about \$100–300 a month**, or roughly **\$0.05–0.15 per company**.

Total running cost once live: about \$130–365 a month

Claude API \$100–300, database \$25, server \$7–20, app hosting \$0–20. Everything else uses free tiers or the client's existing tools.

What changes the bill

Countries and sources watched, pages read, emails drafted, and the model mix. Double the volume, roughly double the API cost.

How we keep it under control

A monthly spending cap and alerts on the API account, a budget per run in our code, and cost per lead on the dashboard.

Disclaimer: vendor list prices were checked in September 2026 and exclude tax. Vendors can change prices, plan limits and usage allowances at any time. All figures are planning estimates, confirmed after the Phase 0 baseline.

16. Risks and how we handle them

The biggest risk is not the AI. It is contact-data coverage in some countries, and email deliverability. Both are handled with measured gates, not promises.

Risk	Likelihood	What we do
Free sources miss some contacts, especially mid-level buyers	High	In-house finder first (company sites, press releases, email-format checks); measure the fill rate in weeks 7–8 on the client's real target list and add a paid source only if the gap is costing deals
Outreach damages the client's main email domain	Medium	Send only from separate sending domains, warmed up over 3–4 weeks; bounce and complaint limits pause campaigns automatically
AI extracts a wrong contractor or project	Medium	Quote-backed facts, the test set run before every release, low-confidence items sent to human review
Tender portals require registration or block automated access	Medium	Use the client's supplier registrations where they have them; otherwise a licensed project database; no scraping against a portal's terms
The client has no Volza account	Medium	Validation falls back to web evidence (past awards, supplier lists); Volza plugs in later through the same interface
Data protection (UAE PDPL, EU GDPR and other local laws)	Medium	Business-contact data only, a lawful-basis record per country (the UAE PDPL defaults to consent; the Saudi PDPL needs consent or prior interaction for marketing), opt-out on every email, a suppression list, retention limits, a data-processing register
Scope growth ("add 6sense features")	Medium	Fixed phase scopes with acceptance tests; new layers become change requests

17. Decisions needed and next steps

Five decisions from the client let us fix the scope and price within a week.

- ☐ Which CRM, if any, the sales team uses today (decides option A or B in section 10)
- ☐ Which products and markets to target first, e.g. line pipe and plates for pipeline projects in the first three target countries
- ☐ Which tools the client already pays for (e.g. Volza, a CRM, Microsoft 365 or Google Workspace) that we can connect at no extra cost
- ☐ Which mailbox and sending domains outreach should use, and who approves emails
- ☐ Scope: the first three stages only, or the full four-stage build

Proposed next step: a two-week Phase 0 (secure and stabilise, plus the test set). It fixes the critical issues immediately and gives both sides real accuracy numbers before committing to the full build.

Sources

Checked in September 2026. Free tools and hosting: [GDELT](#), [Brave Search API](#), [Google Custom Search](#), [Crawl4AI](#), [Crawlee](#), [Jina Reader](#), [Reacher](#), [Hunter](#), [Leadfeeder](#), [Zoho CRM](#), [HubSpot CRM](#), [Supabase billing](#), [Supabase pausing](#), [Supabase Queues](#), [Vercel fair use](#), [Hetzner](#), [Sentry](#), [UN Comtrade](#), [Gemini API terms](#). AI tools: [Claude plans](#), [ChatGPT and Codex plans](#), [Claude Batch API](#). Market data and law: [Tadawul announcement example](#), [Tadawul disclosure rules](#), [OFAC SDN](#), [UN sanctions list](#), [UAE local list](#), [UAE PDPL](#), [Saudi PDPL](#), [Etimad regulations](#). Existing tools and benchmarks: [6sense on Vendr](#), [ZoomInfo on Vendr](#), [Bombora pricing](#), [Bombora co-op](#), [6sense intent data](#), [Instantly benchmark report](#), [Google sender rules](#), [Yahoo sender rules](#), [AWS](#), [Cloudflare](#), [Fastly](#). Claude model prices are from Anthropic's API rate card. Code findings are from our audit of the prototype repository on 24 Sep 2026.